Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/28/2005
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Project Title:	FUNDAMENTAL EVALUATION OF WATER MIST FOR FIRE SAFETY IN SPACECRAFT AND EXTRATERRESTRIAL ENVIRONMENTS		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80401	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	03/03/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Delplanque, Jean Pierre (Colorado School of Mines) McKinnon, Julian Thomas (Colorado School of Mines)		
Grant/Contract No.:	NNC04AA13A		
Performance Goal No.:			
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Task Description:	This investigation is intended to provide an integrated suite of tasks that involve both experimental studies and detailed modeling of fine water mist in fire suppression scenarios relevant to spaceflight needs. All research on water mist will be done in comparison to a baseline of CO2, the suppression agent of choice onboard the ISS and the most abundant component in the Martian atmosphere. The research plan is based on a hierarchical model. First, a study of simplified scenarios is performed at a detailed level. Second, the critical elements of the detailed model are extracted and then encapsulated in a reduced-order manner for inclusion in the next higher-level scenario. The process is continued until a robust understanding of fire suppression at the level of an ISS rack is obtained. The experiments will be performed in both normal and reduced gravity for an accurate validation of the numerical model and to provide observations of fire suppression behavior under extraterrestrial conditions. The overall goal of this fundamental study and the resulting numerical code, validated with experiments, is to evaluate the role of water mist in the next generation of fire safety systems on spacecraft and habitation modules on the surface of other planets. This project will simultaneously satisfy two needs – scientific and NASA programmatic. At the scientific level, the combined experiment/modeling program will lead to a basic understanding of water mist and CO2 fire suppression in realistic fire scenarios (e.g., wire bundles, burning cloth, etc). A determination will be made of the interaction of reduced oxygen concentration and lowered temperature with smoldering rates and smoke generation. This basic understanding will advance the field of fire suppression for both spacecraft and terrestrial applications. At the NASA programmatic level, a series of sub-models (fire source, suppression agent generation and distribution, and radiation models) will be developed resulting on an integrated model that can be employed by NASA as design aids for spaceflight hardware. Deliverables will also fall into scientific and programmatic categories. In the scientific category there will be studies in the form of peer-reviewed journal articles on the basic understanding gained in the sub-models and the issues resolved in model integration. In the programmatic category there will be software modules at the sub-model and integrated model level, and data that will be available to investigators of reduced gravity flame spread, and fire growth and suppression.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results on the effectiveness of the various suppressant agents used in this project will serve to determine an optimum fire-fighting technique for space and for Earth applications. While the main emphasis is on fighting fires on spacecraft, the results will be generalized for all gravity conditions, including Earth's gravity. The software code developed will serve as a tool for designing fire suppression systems for aircraft, ships, museums, libraries, and a variety of enclosed spaces.
Task Progress:	The experimental study of water mist dispersion and the various fire scenarios and fire suppression methods will be carried out in a test apparatus that is modeled after the Space Shuttle Middeck Locker (SSML). To replicate a fire suppression scenario under these conditions, we have started the construction of the Middeck Locker Fire Testing Apparatus (MLFTA), which is built of aluminum and tempered glass and has approximate dimensions of 17.3” x 10.0” x 20.3”. The top panel of the apparatus is transparent to allow video data collection and an adjustable ventilation system provides airflow and purging of combustion gases. In this pseudo-locker we will test a variety of suppression methods, such as water-mist, carbon dioxide, and nitrogen with single and multiple injection points. These injection methods cover the current ISS fire suppression method of a portable hand-held fire extinguisher spraying through a port in a rack and also possible next-generation spacecraft components that may have a multi-point suppression delivery system built into the design. We will also test hybrid suppression methods such as water mist with a CO2 or N2 atomizing fluid. At first, experiments will be done in our laboratory in normal gravity where we have the greatest access to instrumentation (e.g., gas chromatography) and the experiments can be carried out with greater repetition rates. The normal gravity experiments will provide preliminary heat release rate input for the fire source numerical sub-model and will also be used to identify worst-case burning materials and fire configurations. In the past few months, we have conducted the first experiments with polyethylene-insulated cable to measure ignition energies and times, burning behavior, flame spread rates, and heat-release rates under upward, downward, and horizontal configurations. We plan to continue this type of testing to collect information on all the different materials that will be burned and suppressed in this investigation. In anticipation of the experiments that will be conducted in reduced gravity, we met with members of the Reduced Gravity Office at NASA GRC to discuss the requirements for rig construction, safety practices, documentation, and possible flight dates for this experiment. The long-term modeling effort of this project consists of developing detailed sub-models of the fire source, the suppression agent generation and distribution, and the radiative shielding of the suppression agent. These sub-models will then be integrated into a high-fidelity, fire-suppression model. Finally, a reduced order model will be developed to minimize the computational requirements, yet retain the simulation capabilities of the original formulation. To date, we have obtained preliminary results in the development of the polydispersed mist model, where water mist injection characteristics include the number of injectors and their position in the domain considered, the mass flow rate injected (which may be time-dependent), the injection orifice size, the droplet size and velocity distribution, and the spray cone-angle. These preliminary numerical results provide interesting insight regarding the mist evaporation dynamics that is directly relevant to the issue of droplet penetration in the presence of obstacles.
Bibliography Type:	Description: (Last Updated: 03/21/2008)
Papers from Meeting Proceedings	Delplanque JP, Abbud-Madrid A, McKinnon JT, Lewis SJ, Watson D. "Feasibility Study of Water Mist for Spacecraft Fire Suppression." Halon Options Technical Working Conference (HOTWC-04), May 2004. Proceedings of the Halon Options Technical Working Conference (HOTWC-04), May 2004. , May-2004
Papers from Meeting Proceedings	Johnson SB, Lewis S, Delplanque JP. "Droplet-Resolved Dilute Spray Evaporation Model." Heat Transfer/Fluids Engineering Summer Conference, Charlotte, North Carolina, July 11-15, 2004. Proceedings of Heat Transfer/Fluids Engineering Summer Conference, Charlotte, North Carolina, July 11-15, 2004. ASME Publication HT-FED2004-56898, 2004. , Jul-2004
Papers from Meeting Proceedings	Lewis SJ, Delplanque JP. "Fire Suppression in Microgravity: Dynamics of a Polydispersed Water Mist around an Obstacle." 2004 ASME International Mechanical Engineering Congress, November 2004. Proceedings of 2004 ASME International Mechanical Engineering Congress, November 2004. , Nov-2004

Presentation	Lewis, S., Delplanque, J. P., Modak, A., and Kee, R. J. "Behavior of Polydispersed Water Mist during Simulated Fire Suppression Events" Spring Meeting of the Western States Sections of the Combustion Institute, Davis, CA Mar-2004